SSC MTS 2024 question paper with answers

17 October 2024 · Shift 3

80 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 10 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 14 questions

Q.1

Amithab has scored an average of 72 runs in his six interschool matches. In the next match, he scored 96 runs. What is his revised average score?

(A) 75.428
(B) 74
(C) 75
(D) 75.3

Q.2

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 7% per year rate of interest for 3 years.

(A) 570
(B) 590
(C) 630
(D) 670

Q.3

The average weight of a group of 8 people is 70 kg. After a new person joins the group, the average weight becomes 72 kg. What is the weight (in kg) of the new person?

(A) 86
(B) 87
(C) 89
(D) 88

Q.4

A person invested a certain sum of money at 6% per annum simple interest and another sum at 8% per annum simple interest at the same time. After 4 years, he received interest ₹1,560 from both sums. One-fifth of the first sum is equal to one-sixth of the second sum. Find the total sum invested.

(A) ₹6,000
(B) ₹5,000
(C) ₹5,500
(D) ₹4,500

Q.5

The side of a square exceeds the side of another square by 3 cm and the sum of the areas of the two squares is 225 cm^2 . Find the sum of the perimeter (in cm) of both squares.

(A) 74
(B) 64
(C) 94
(D) 84

Q.7

A well with an internal diameter 10 m is dug to a depth of 10 m. Earth taken out of it is spread all around the well to a width of 12 m to form an embankment. What is the height (in m, rounded off to 2 decimal places) of the embankment?

(A) 0.88
(B) 1.02
(C) 1.12
(D) 0.95

Q.11
What sum (in ₹) will earn a simple interest of ₹360 in 2 years at 8% per year rate of interest?

(A) 2250
(B) 1125
(C) 3000
(D) 4500

Q.12
Raima saves 14% from her income of ₹35,000. How much is her expenditure (in ₹)?

(A) 30100
(B) 32000
(C) 28600
(D) 34000

Q.13
A thief is noticed by a policeman from a distance of 150 m. The thief starts running, and the policeman chases him. The thief and the policeman run at the rate of 8 m/s and 10 m/s, respectively. Find the distance travelled by the thief before being caught.

(A) 600 m
(B) 700 m
(C) 500 m
(D) 550 m

Q.14
A dealer marked the price of an item 30% above the cost price. He allowed two successive discounts of 15% and 20% to a particular customer. As a result, he incurred a loss of ₹696. At what price did he sell the item to the customer?

(A) ₹5,304
(B) ₹6,205
(C) ₹4,904
(D) ₹7,805

Q.15
The average of 70 values is 40. If each value is multiplied by 20, what will be the changed average?

(A) 60
(B) 1400
(C) 90
(D) 800

Q.16
If the radius of the base of a right circular cylinder is reduced to one-third, keeping the height same, what is the ratio of the volume of the reduced cylinder to that of the original?

(A) 2 : 4
(B) 1 : 9
(C) 1 : 2
(D) 2 : 3

Q.17
If two kinds of resins costing ₹160 per kg and ₹210 per kg are mixed in the ratio of 2 : 3, then find the cost (in ₹) of the mixture per kg.

(A) 185
(B) 190
(C) 180
(D) 200

Q.20
A length of a violin string varies inversely with respect to the frequency of its vibrations. The frequency of a 14-inch-long violin string is 450 cycles per second. Find the frequency of a 12-inch violin string.

(A) 510 cycles per second
(B) 525 cycles per second
(C) 520 cycles per second
(D) 515 cycles per second

Reasoning Ability and Problem Solving 16 questions

Q.1

KB 9 is related to ED 2 in a certain way. In the same way, XF 11 is related to RH 4. To which of the following is UH 8 related, following the same logic?

(A) OK 1
(B) OK 3
(C) OJ 3
(D) OJ 1

Q.2

326 is related to 335 following a certain logic. Following the same logic, 419 is related to 428. To which of the following is 566 related, following the same logic? (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding / subtracting / multiplying, etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 599
(B) 582
(C) 575
(D) 571

Q.4

In a certain code language, 'TREE' is coded as 'POCD' and 'FOOT' is coded as 'BLMS'. How will 'MIND' be coded in the same language?

(A) JEKD
(B) IFLC
(C) JFLD
(D) IEKD

Q.5

A, B, C, D, E, F and G are sitting around a circular table facing the centre(but not necessarily in the same order). F is sitting third to the left of A. A is sitting second to the left of G. G is sitting to the immediate right of E. B is sitting to the immediate left of D. What is the position of F with respect to E?

(A) Second to the left
(B) Immediate left
(C) Immediate right
(D) Third to the right

Q.6

In a certain code language, 'apple pen card' is coded as '136', 'doctor no apple' is coded as '243' and 'card is blocked' is coded as '695'. How is 'apple' coded in the given code language?

(A) 6
(B) 4
(C) 3
(D) 1

Q.8

What should come in place of the question mark (?) in the given series based on the English alphabetical order? PKQ ,OJP, NIO ,MHN, ?

(A) NGL
(B) LGN
(C) LGM
(D) MGL

Q.9

What will come in place of the question mark (?) of the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 45 × 3 + 9 – 18 ÷ 1 = ?

(A) 12
(B) 15
(C) 24
(D) 6

Q.10

What should come in place of the question mark (?) in the following series? 255, 248, 241, 234 ,227, ?

(A) 207
(B) 221
(C) 220
(D) 217

Q.11

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: All doors are locks. Some locks are handles, Conclusions: (I) All handles are locks. (II) No door is a handle.

(A) Only conclusion I follows
(B) Only conclusion II follows
(C) Neither conclusion I nor II follows
(D) Both conclusions I and II follow

Q.12

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statement(s). Statements: All cabbages are roses. All cabbages are tulips. Conclusions: (I) At least some tulips are roses. (II) All roses are cabbages.

(A) Only conclusion (I) follows
(B) Only conclusion (II) follows
(C) Both conclusions (I) and (II) follow
(D) Neither conclusion (I) nor (II) follows

Q.13

Which two signs should be interchanged to make the given equation correct? $11 \times 7 - 48 + 6 \div 37 = 106$

(A) ÷ and +
(B) – and +
(C) + and ×
(D) – and ×

Q.14

In a certain code language, ‘LIKE’ is coded as ‘4682’ and ‘BEAM’ is coded as ‘1358’.What is the code for ‘E’ in the given code language?

(A) 2
(B) 5
(C) 3
(D) 8

Q.15

LQ87 is related to VA55 in a certain way. In the same way, QV59 is related to AF27. To which of the following is BG49 related, following the same logic?

(A) LQ19
(B) LQ18
(C) QL17
(D) LQ17

Q.17

In a code language, ‘I want food’ is written as ' sickota dongre dujubu’, ‘I want drink’ is written as ‘bluto sickota dujubu', ‘food and drink’ is written as ‘dongre peppin bluto’, then what is the code for the word 'peppin' in this language?

(A) I
(B) drink
(C) food
(D) and

Q.18

T, N, S, P, R and M are sitting around a circular table facing the centre (but not necessarily in the same order). S is sitting second to the left of T. Only two people are sitting between S and R. M is not the immediate neighbour of S. P is sitting second to the right of R. How many people are sitting between N and M, when counted from the right of N?

(A) Three
(B) Two
(C) Four
(D) One

Q.19

In the following number-pairs, the second number is obtained by applying certain mathematical operations to the first number. Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) 11, 66; 26, 156

(A) 18, 108
(B) 20, 130
(C) 26, 152
(D) 15, 75

General Awareness 25 questions

Q.1

The weaker warm storms that grow stronger over time are finally converted into huge storms or hurricanes. Which of the following climate regions is the origin of such hurricanes?

(A) Semi-arid climate region
(B) Continental climate region
(C) Tropical climate region
(D) Arid climate region

Q.2

Which type of loan is supported by MUDRA?

(A) Primary loan
(B) Real estate loan
(C) Corporate loan
(D) Non-farm loan

Q.3

India launched the 'National Social Assistance Programme' in:

(A) 1995
(B) 1990
(C) 1985
(D) 2000

Q.4

The Iron Age site named Tekwada is located in _____.

(A) Tamil Nadu
(B) Maharashtra
(C) Madhya Pradesh
(D) Andhra Pradesh

Q.5

In which of the following states is the Bissu Mela organised to thank Santoora Devi for the good harvest?

(A) Punjab
(B) Himachal Pradesh
(C) Uttarakhand
(D) Haryana

Q.6

In Union Budget 2023, a sub-scheme of which of the following Yojanas has been proposed with the investment of ₹6,000 crore to enable activities of fishermen, fish vendors and micro and small enterprises?

(A) PM Mudra Yojana
(B) PM Ujjwala Yojana
(C) PM Fasal Bima Yojana
(D) PM Matsya Sampada Yojana

Q.7

How many chances does a high jumper get for overcoming a certain height?

(A) Six chances
(B) Three chances
(C) Two chances
(D) Four chances

Q.8

In Harappan civilisation, cities were planned with the streets generally oriented along the _____ directions.

(A) incognito
(B) cardinal
(C) ordinal
(D) anonymous

Q.9

In june 2022, Eknath Sambhaji Shinde became the Chief Minister of which of the following states?

(A) Gujarat
(B) Haryana
(C) Rajasthan
(D) Maharashtra

Q.10

Which Census decade recorded the highest population growth rate (in terms of percentage) in India?

(A) 1971-1981
(B) 1941-1951
(C) 1961-1971
(D) 2001-2011

Q.11

To which of the following groups do whales belong?

(A) Amphibia
(B) Arthropoda
(C) Mammalia
(D) Pisces

Q.12

Who among the following received the Padma Shri 2021 for Panthi folk dance?

(A) Radhe Shyam Barle
(B) Bhuri Bai
(C) Jagdish Chaudhary
(D) Gulfam Ahmed

Q.13

The term 'Aus', 'Aman' and 'Boro' are associated with which crop in India?

(A) Jute
(B) Wheat
(C) Paddy
(D) Maize

Q.14

What is the percentage of 's' character in the carbon atom of acetylene?

(A) 50%
(B) 33.33%
(C) 25%
(D) 75%

Q.15

Which of the following is the correct statement about the Rajya Sabha?

(A) It is known as the Council of the States.
(B) It is known as the lower house of the parliament.
(C) All the members are directly elected by the people.
(D) The term of the members is for 5 years.

Q.16

Anorexia Nervosa is an eating disorder characterised by _____. a) an abnormally heavy body weight b) an intense fear of losing weight c) a distorted perception of weight d) trying to lose weight by exercising excessively

(A) a and b
(B) c and d
(C) a and c
(D) b and c

Q.17
Buddhadev Dasgupta was a famous instrumentalist of which of the following instrument?

(A) Sitar
(B) Surbahar
(C) Santoor
(D) Sarod

Q.18
Which of the following Mughal emperors issued silver coins in the honour of his wife Mehrunnisa?

(A) Jahangir
(B) Aurangzeb
(C) Akbar
(D) Humayun

Q.19
The author of the Hindi novel ‘Kohre Mein Kaid Rang’ is:

(A) Govind Mishra
(B) Kamleshwar
(C) Uday Prakash
(D) Chitra Mudgal

Q.20
Who among the following founded the 'Satyashodhak Samaj' in 1873?

(A) Gopal Ganesh Agarkar
(B) Jyotiba Phule
(C) Vinoba Bhave
(D) Vinayak Damodar Savarkar

Q.21
Rabindranath Tagore referred to _____ as Nritya Samragini, which translates to 'the empress of dance'.

(A) Mrinalini Sarabhai
(B) Sitara Devi
(C) Gangu Bai
(D) Rukmini Arundale

Q.22
Which of the following Fundamental Rights entails trafficking of human beings and forced labour?

(A) Cultural and Educational Rights
(B) Right to Equality
(C) Rights against Exploitation
(D) Right to Freedom

Q.23
Who among the following founded the Asiatic Society of Bengal?

(A) Henry Colebrook
(B) William Jones
(C) Robert Clive
(D) Warren Hastings

Q.24
Ujjwala was the mascot of which national sports event held in 2023?

(A) Khelo India University Games 2022
(B) Khelo India Youth Games 2023
(C) National Games 2023
(D) Khelo India Para Games 2023

Q.25
Which of the following is NOT a physiographic division of India?

(A) The Tibetan Plateau
(B) The Northern Plains
(C) The Peninsular Plateau
(D) The Himalayan Mountains

English Language and Comprehension 25 questions

Q.1
Select the most appropriate to substitute the underlined word in the given sentence. Rohit asked Nita what was the most “intriguing” book she has read recently.

(A) subdued
(B) encircled
(C) tedious
(D) arraigned

Q.2

Select the INCORRECTLY spelt word..

(A) Conscious
(B) Receive
(C) Occurrance
(D) Believe

Q.3

Select the most appropriate option to fill in the blank. The ancient Greeks and pagans used to worship natural _____ as Gods and Goddesses.

(A) elements
(B) impulses
(C) instincts
(D) mixtures

Q.4

Select the most appropriate idiom that can substitute the underlined segment in the given sentence. Most women are “facing the same problem” when it comes to job satisfaction because of the pay parity in institutions.

(A) in the same boat
(B) comparing apples to oranges
(C) armchair experts
(D) a drop in the ocean

Q.5

Parts of the following sentence have been given as options. Select the option that contains an error. Rose painted the most beautifullest picture ever at the art class today!

(A) art class today!
(B) the most beautifullest
(C) picture ever at the
(D) Rose painted

Q.6

Correct the underlined part of the sentence. This mango is “bitterer”.

(A) most bitter
(B) bitter
(C) bitterest
(D) more bitter

Q.7

Select the INCORRECTLY spelt word.

(A) Inflamous
(B) Kerbstone
(C) Arrangement
(D) Subjugator

Q.8

Select the option that can be used as a one-word substitute for the given group of words. one who does something for pleasure

(A) Atheist
(B) Hedonist
(C) Aggressor
(D) Anarchist

Q.9

Select the most appropriate ANTONYM of the underlined word. Raju is “lazy” and smart, but he does not attend the classes.

(A) lax
(B) industrious
(C) genius
(D) snobbish

Q.10

Select the most appropriate option to fill in the blank. He pressed the _____ just in time to avoid the accident.

(A) bike
(B) brake
(C) beak
(D) break

Q.11

Select the most appropriate ANTONYM of the underlined word in the given sentence. Even if an unlaced shoe is spotless, it is not “neat”.

(A) soiled
(B) forged
(C) agile
(D) requisite

Q.12

Select the grammatically correct sentence..

(A) The traffic on the highway was heavy during an evening.
(B) A traffic on the highway was heavy during an evening.
(C) The traffic on a highway was heavy during an evening.
(D) The traffic on the highway was heavy during the evening.

Q.13

Select the most appropriate option that can substitute the underlined segment in the given sentence. My brother asked me, "Can I exchange “my phone from you”?"

(A) my phone with yours
(B) my phone since yours
(C) my phone off yours
(D) my phone within yours

Q.14

Select the most appropriate synonym of the underlined word in the given sentence. Any declarations or practices of religious “antipathy” against ethnic minorities in a democratic country should not be tolerated.

(A) Enmity
(B) Devaluation
(C) Conformity
(D) Passivity

Q.15

Select the option that expresses the given sentence in future perfect tense form. I hope you wash all the clothes before you come into the house again.

(A) I hope you will have wash all the clothes before you come into the house again.
(B) I hope you shall wash all the clothes before you come into the house again
(C) I hope you will have washed all the clothes before you come into the house again.
(D) I hope you will be washing all the clothes before you come into the house again.

Q.16

Select the INCORRECTLY spelt word...

(A) Battle
(B) Comfortable
(C) Depresed
(D) Cheerful

Q.17

Select the most appropriate option that can substitute the underlined segment in the given sentence. Everyone disliked the new CEO, Kushal, but nobody is ready to “put themselves in dangerous situation” by telling him their feelings.

(A) bell the cat
(B) sit on the fence
(C) pull a long face
(D) rule the roost

Q.18

Select the grammatically correct sentence.

(A) The intrepid explorer stumbled upon the uncharted cavern deep within the heart of the rainforest.
(B) An intrepid explorer stumbled upon an uncharted cavern deep within the heart of the rainforest.
(C) The intrepid explorer stumbled upon an uncharted cavern deep within a heart of the rainforest.
(D) An intrepid explorer stumbled upon the uncharted cavern deep within a heart of the rainforest.

Q.19

Read the sentence given below and identify if there is an error in it. The error can be in any segment of the sentence which are marked A, B, C, and D. Select option D if there is 'No error'. (A) She sings / (B) the song / (C) beautiful in the party / (D) No error.

(A) A
(B) B
(C) D
(D) C

Q.20

Select the most appropriate option to fill in the blank. I am calling to thank you _____ the present you sent.

(A) of
(B) in
(C) for
(D) with

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Newspapers are like mirrors (1)_____ the diverse facets of society, capturing the voices, stories, and events that define our shared human experience. They serve as a forum (2)_____ dialogue, debate, and discussion, amplifying voices from all walks of life and fostering (3)_____ sense of community. Beyond reporting news, newspapers also celebrate achievements, highlight cultural (4)_____, and showcase human resilience in the face of adversity. They are archives of history, chronicling the triumphs and trials of generations, and serving as a reminder of how far we have come and how much further we can go. In an age of digital transformation, newspapers continue to (5)_____, adapting to new mediums while upholding the core values of journalism: accuracy, integrity, and accountability.

Select the most appropriate option to fill in blank number 1.

(A) reflects
(B) reflect
(C) reflected
(D) reflecting

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Newspapers are like mirrors (1)_____ the diverse facets of society, capturing the voices, stories, and events that define our shared human experience. They serve as a forum (2)_____ dialogue, debate, and discussion, amplifying voices from all walks of life and fostering (3)_____ sense of community. Beyond reporting news, newspapers also celebrate achievements, highlight cultural (4)_____, and showcase human resilience in the face of adversity. They are archives of history, chronicling the triumphs and trials of generations, and serving as a reminder of how far we have come and how much further we can go. In an age of digital transformation, newspapers continue to (5)_____, adapting to new mediums while upholding the core values of journalism: accuracy, integrity, and accountability.

Select the most appropriate option to fill in blank number 2.

(A) on
(B) into
(C) by
(D) for

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Newspapers are like mirrors (1)_____ the diverse facets of society, capturing the voices, stories, and events that define our shared human experience. They serve as a forum (2)_____ dialogue, debate, and discussion, amplifying voices from all walks of life and fostering (3)_____ sense of community. Beyond reporting news, newspapers also celebrate achievements, highlight cultural (4)_____, and showcase human resilience in the face of adversity. They are archives of history, chronicling the triumphs and trials of generations, and serving as a reminder of how far we have come and how much further we can go. In an age of digital transformation, newspapers continue to (5)_____, adapting to new mediums while upholding the core values of journalism: accuracy, integrity, and accountability.

Select the most appropriate option to fill in blank number 3.

(A) the
(B) a
(C) No article required
(D) an

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Newspapers are like mirrors (1)_____ the diverse facets of society, capturing the voices, stories, and events that define our shared human experience. They serve as a forum (2)_____ dialogue, debate, and discussion, amplifying voices from all walks of life and fostering (3)_____ sense of community. Beyond reporting news, newspapers also celebrate achievements, highlight cultural (4)_____, and showcase human resilience in the face of adversity. They are archives of history, chronicling the triumphs and trials of generations, and serving as a reminder of how far we have come and how much further we can go. In an age of digital transformation, newspapers continue to (5)_____, adapting to new mediums while upholding the core values of journalism: accuracy, integrity, and accountability.

Select the most appropriate option to fill in blank number 4.

(A) richness
(B) richer
(C) richest
(D) richly

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Newspapers are like mirrors (1)_____ the diverse facets of society, capturing the voices, stories, and events that define our shared human experience. They serve as a forum (2)_____ dialogue, debate, and discussion, amplifying voices from all walks of life and fostering (3)_____ sense of community. Beyond reporting news, newspapers also celebrate achievements, highlight cultural (4)_____, and showcase human resilience in the face of adversity. They are archives of history, chronicling the triumphs and trials of generations, and serving as a reminder of how far we have come and how much further we can go. In an age of digital transformation, newspapers continue to (5)_____, adapting to new mediums while upholding the core values of journalism: accuracy, integrity, and accountability.

Select the most appropriate option to fill in blank number 5.

(A) involve
(B) dissolve
(C) revolve
(D) evolve

Answer key

Numerical and Mathematical Ability

Q.1 **(A)**	Q.2 **(C)**	Q.3 **(D)**	Q.4 **(C)**	Q.5 **(D)**	Q.7 **(D)**
Q.11 **(A)**	Q.12 **(A)**	Q.13 **(A)**	Q.14 **(A)**	Q.15 **(D)**	Q.16 **(B)**
Q.17 **(B)**	Q.20 **(B)**				

Reasoning Ability and Problem Solving

Q.1 **(D)**	Q.2 **(C)**	Q.4 **(B)**	Q.5 **(D)**	Q.6 **(C)**	Q.8 **(C)**
Q.9 **(C)**	Q.10 **(C)**	Q.11 **(C)**	Q.12 **(A)**	Q.13 **(A)**	Q.14 **(D)**
Q.15 **(D)**	Q.17 **(D)**	Q.18 **(B)**	Q.19 **(A)**		

General Awareness

Q.1 **(C)**	Q.2 **(D)**	Q.3 **(A)**	Q.4 **(B)**	Q.5 **(C)**	Q.6 **(D)**
Q.7 **(B)**	Q.8 **(B)**	Q.9 **(D)**	Q.10 **(C)**	Q.11 **(C)**	Q.12 **(A)**
Q.13 **(C)**	Q.14 **(A)**	Q.15 **(A)**	Q.16 **(B)**	Q.17 **(D)**	Q.18 **(A)**
Q.19 **(A)**	Q.20 **(B)**	Q.21 **(B)**	Q.22 **(C)**	Q.23 **(B)**	Q.24 **(D)**
Q.25 **(A)**					

English Language and Comprehension

Q.1 **(C)**	Q.2 **(C)**	Q.3 **(A)**	Q.4 **(A)**	Q.5 **(B)**	Q.6 **(B)**
Q.7 **(A)**	Q.8 **(B)**	Q.9 **(B)**	Q.10 **(B)**	Q.11 **(A)**	Q.12 **(D)**
Q.13 **(A)**	Q.14 **(A)**	Q.15 **(C)**	Q.16 **(C)**	Q.17 **(A)**	Q.18 **(B)**
Q.19 **(D)**	Q.20 **(C)**	Q.21 **(D)**	Q.22 **(D)**	Q.23 **(B)**	Q.24 **(A)**
Q.25 **(D)**					

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (A) 75.428

Six matches at 72 make 6 × 72 = 432 runs; with 96 more the total is 528 over 7 matches: 528/7 = 75.428...

Q.2 · Answer (C) 630

SI = 3000 × 7 × 3/100 = ₹630.

Q.3 · Answer (D) 88

The total rises from 8 × 70 = 560 kg to 9 × 72 = 648 kg, so the new person weighs 648 − 560 = 88 kg.

Q.4 · Answer (C) ₹5,500

One-fifth of the first sum equals one-sixth of the second, so the sums are 5k and 6k. A year's interest is 0.06 × 5k + 0.08 × 6k = 0.78k, and 4 × 0.78k = 1,560 gives k = 500. Total = 11k = ₹5,500.

Q.5 · Answer (D) 84

With sides x + 3 and x: $(x + 3)^2 + x^2 = 225$, so $x^2 + 3x − 108 = 0$ and x = 9. The sides are 12 cm and 9 cm, and the perimeters add up to 4 × (12 + 9) = 84 cm.

Q.7 · Answer (D) 0.95

Earth dug out = $\pi \times 5^2 \times 10 = 250\pi$ m³. The embankment is a ring from radius 5 m to 5 + 12 = 17 m, of area $\pi(17^2 − 5^2) = 264\pi$ m². Height = 250π/264π = 0.947 ≈ 0.95 m.

Q.11 · Answer (A) 2250

P = SI × 100/(R × T) = 360 × 100/(8 × 2) = ₹2,250.

Q.12 · Answer (A) 30100

She spends 100 − 14 = 86% of her income: 0.86 × 35,000 = ₹30,100.

Q.13 · Answer (A) 600 m

The policeman gains 10 − 8 = 2 m/s on the thief, so he closes the 150 m gap in 75 seconds. In that time the thief runs 8 × 75 = 600 m.

Q.14 · Answer (A) ₹5,304

With cost price C, the selling price is 1.3C × 0.85 × 0.8 = 0.884C, a loss of 0.116C = ₹696. So C = ₹6,000 and the selling price is 0.884 × 6,000 = ₹5,304.

Q.15 · Answer (D) 800

Multiplying every value by 20 multiplies the average by 20: 40 × 20 = 800.

Q.16 · Answer (B) 1 : 9

$V = \pi r^2 h$, so with the height unchanged the volume varies as r^2: $(1/3)^2 = 1/9$, a ratio of 1 : 9.

Q.17 · Answer (B) 190

(2 × 160 + 3 × 210)/5 = (320 + 630)/5 = 950/5 = ₹190 per kg.

Q.20 · Answer (B) 525 cycles per second

Length × frequency stays the same: 14 × 450 = 6,300, so a 12-inch string vibrates at 6,300/12 = 525 cycles per second.

Reasoning Ability and Problem Solving

Q.1 · Answer (D) OJ 1

The first letter moves 6 back, the second 2 forward and the number falls by 7: K → E, B → D, 9 → 2 and X → R, F → H, 11 → 4. So U → O, H → J and 8 → 1: OJ 1.

Q.2 · Answer (C) 575

The second number is the first plus 9: 326 + 9 = 335 and 419 + 9 = 428. Likewise 566 + 9 = 575.

Q.4 · Answer (B) IFLC
The letters move back 4, 3, 2 and 1 places: TREE → POCD and FOOT → BLMS. So M → I, I → F, N → L and D → C: IFLC.

Q.5 · Answer (D) Third to the right
Going round the table to each person's left from E: E, A, D, B, F, C, G. This fits every clue (G to E's immediate right, A second to G's left, F third to A's left, B to D's immediate left), and F is three places to E's right.

Q.6 · Answer (C) 3
'apple' is the only word in both the first and second sentences and 3 the only digit in both their codes, so apple = 3.

Q.8 · Answer (C) LGM
Every letter moves one place back each time: P, O, N, M, L; K, J, I, H, G; Q, P, O, N, M. The next cluster is LGM.

Q.9 · Answer (C) 24
After interchanging the signs: 45 ÷ 3 – 9 + 18 × 1 = 15 – 9 + 18 = 24.

Q.10 · Answer (C) 220
Each term is 7 less than the one before: 227 – 7 = 220.

Q.11 · Answer (C) Neither conclusion I nor II follows
Only some locks are said to be handles, and nothing is said of handles that are not locks, so I does not follow. The doors may or may not be among the locks that are handles, so II does not follow either.

Q.12 · Answer (A) Only conclusion (I) follows
Every cabbage is both a rose and a tulip, so some tulips are roses (I). Nothing says every rose is a cabbage, so II does not follow.

Q.13 · Answer (A) ÷ and +
Interchanging ÷ and +: 11 × 7 – 48 ÷ 6 + 37 = 77 – 8 + 37 = 106.

Q.14 · Answer (D) 8
E is the only letter in both words and 8 the only digit in both codes, so E = 8.

Q.15 · Answer (D) LQ17
Both letters move 10 places forward (wrapping round after Z) and the number falls by 32: L → V, Q → A, 87 → 55 and Q → A, V → F, 59 → 27. So B → L, G → Q and 49 → 17: LQ17.

Q.17 · Answer (D) and
'I want' is in the first two sentences, so 'sickota' and 'dujubu' stand for 'I' and 'want'; that leaves food = dongre and drink = bluto. The code left over in the third sentence, 'peppin', is 'and'.

Q.18 · Answer (B) Two
Going round the table to each person's left from T: T, N, S, P, M, R (S second to T's left, R opposite S, P second to R's right, M not beside S). Counting from N to the right, T and R come before M: two people.

Q.19 · Answer (A) 18, 108
The second number is 6 times the first: 11 × 6 = 66 and 26 × 6 = 156. Likewise 18 × 6 = 108.

General Awareness

Q.1 · Answer (C) Tropical climate region
Hurricanes are tropical cyclones: they form over warm tropical seas, whose surface water above about 27 °C supplies the heat and moisture that let a weak storm grow.

Q.2 · Answer (D) Non-farm loan
MUDRA (Micro Units Development and Refinance Agency), set up in 2015, supports loans to non-corporate, non-farm small and micro enterprises under the Pradhan Mantri Mudra Yojana.

Q.3 · Answer (A) 1995
The National Social Assistance Programme, with old-age pension, family benefit and maternity benefit schemes, was launched on 15 August 1995.

Q.4 · Answer (B) Maharashtra
SSC's key gives Maharashtra, where Tekwada is one of the excavated sites with Iron Age remains.

Q.5 · Answer (C) Uttarakhand
The Bissu Mela of the Jaunsari people, held in the Jaunsar-Bawar area (Chakrata) of Uttarakhand's Dehradun district, honours the goddess Santoora Devi at harvest time.

Q.6 · Answer (D) PM Matsya Sampada Yojana
The Union Budget 2023-24 announced a new sub-scheme of the PM Matsya Sampada Yojana, with a targeted investment of ₹6,000 crore, for fishermen, fish vendors and micro and small enterprises.

Q.7 · Answer (B) Three chances
A high jumper has three attempts at a height; three failures in a row put the athlete out of the competition.

Q.8 · Answer (B) cardinal
Harappan streets ran roughly north-south and east-west, the cardinal directions, crossing at right angles in a grid.

Q.9 · Answer (D) Maharashtra
Eknath Shinde was sworn in as Chief Minister of Maharashtra on 30 June 2022.

Q.10 · Answer (C) 1961-1971
India's population grew fastest in the 1961-1971 decade, by 24.80%, slightly more than in 1971-1981.

Q.11 · Answer (C) Mammalia
Whales breathe air through lungs, are warm-blooded and suckle their young: they are mammals, not fish.

Q.12 · Answer (A) Radhe Shyam Barle
Radhe Shyam Barle of Chhattisgarh received the Padma Shri in 2021 for Panthi, the folk dance of the Satnami community.

Q.13 · Answer (C) Paddy
In West Bengal farmers grow three crops of rice (paddy) a year, called 'aus', 'aman' and 'boro'.

Q.14 · Answer (A) 50%
Each carbon in acetylene (HC≡CH) is sp hybridised, one s and one p orbital, so it has 50% s character (sp^2 has 33.33%, sp^3 25%).

Q.15 · Answer (A) It is known as the Council of the States.
The Rajya Sabha is the Council of States, Parliament's upper house. Most of its members are elected indirectly by the state legislative assemblies, for six years.

Q.16 · Answer (B) c and d
Anorexia nervosa involves a distorted perception of one's weight and efforts such as excessive exercise to lose weight. Sufferers have an abnormally low (not heavy) body weight and an intense fear of gaining (not losing) weight.

Q.17 · Answer (D) Sarod
Pandit Buddhadev Dasgupta, a disciple of Radhika Mohan Maitra, was one of the leading sarod players of his time.

Q.18 · Answer (A) Jahangir
Mehrunnisa became Jahangir's queen, Nur Jahan, in 1611; Jahangir had coins struck in her name, a rare honour for a Mughal queen.

Q.19 · Answer (A) Govind Mishra
Govind Mishra won the 2008 Sahitya Akademi Award for his Hindi novel 'Kohre Mein Kaid Rang'.

Q.20 · Answer (B) Jyotiba Phule
Jyotirao (Jyotiba) Phule founded the Satyashodhak Samaj in Pune on 24 September 1873 to fight caste discrimination.

Q.21 · Answer (B) Sitara Devi
Rabindranath Tagore gave the Kathak dancer Sitara Devi the title 'Nritya Samragini' after seeing her dance as a girl.

Q.22 · Answer (C) Rights against Exploitation
The Right against Exploitation (Articles 23 and 24) bans trafficking in human beings and forced labour (Article 23) and the employment of children in hazardous work (Article 24).

Q.23 · Answer (B) William Jones
Sir William Jones founded the Asiatic Society in Calcutta on 15 January 1784.

Q.24 · Answer (D) Khelo India Para Games 2023
Ujjwala, a sparrow standing for Delhi, was the mascot of the first Khelo India Para Games, held in New Delhi in December 2023.

Q.25 · Answer (A) The Tibetan Plateau
NCERT divides India into six physiographic divisions: the Himalayan Mountains, the Northern Plains, the Peninsular Plateau, the Indian Desert, the Coastal Plains and the Islands. The Tibetan Plateau lies beyond the Himalayas, in Tibet.

English Language and Comprehension

Q.1 · Answer (C) tedious
The instruction, as SSC printed it, is missing a word (most likely 'ANTONYM'). SSC's key gives 'tedious', dull and tiresome, the opposite of 'intriguing', fascinating; it is also the only option that makes sense in the sentence, and none of the options means the same as 'intriguing'.

Q.2 · Answer (C) Occurrance
The correct spelling is 'occurrence'.

Q.3 · Answer (A) elements
The ancients worshipped natural 'elements', such as the sun, wind, water and fire, as gods and goddesses.

Q.4 · Answer (A) in the same boat
To be 'in the same boat' is to share the same difficult situation with others.

Q.5 · Answer (B) the most beautifullest
'Beautiful' takes 'most' for its superlative: 'the most beautiful picture'. 'Most beautifullest' is a double superlative, and 'beautifullest' is not a word.

Q.6 · Answer (B) bitter
Nothing is compared with the mango, so the positive degree is needed: 'This mango is bitter.'

Q.7 · Answer (A) Inflamous
'Inflamous' is not a word; the spelling is 'infamous', well known for something bad. 'Kerbstone' is the British spelling of 'curbstone'.

Q.8 · Answer (B) Hedonist
A hedonist is one who lives for and pursues pleasure. (One who takes up an activity for pleasure rather than pay is an 'amateur', which is not among the options.)

Q.9 · Answer (B) industrious
'Lazy' means unwilling to work; its opposite is 'industrious', hard-working. 'Lax' is close to a synonym.

Q.10 · Answer (B) brake
A vehicle is stopped with its 'brake'; 'break' sounds the same but means something else.

Q.11 · Answer (A) soiled
'Neat' means clean and tidy; its opposite is 'soiled', made dirty.

Q.12 · Answer (D) The traffic on the highway was heavy during the evening.
The highway and the evening are particular ones, so both take 'the': 'The traffic on the highway was heavy during the evening.'

Q.13 · Answer (A) my phone with yours
Of the options only 'with' goes with 'exchange': 'Can I exchange my phone with yours?' 'Since', 'off' and 'within' do not.

Q.14 · Answer (A) Enmity
'Antipathy' is a deep-seated dislike or hostility: 'enmity'.

Q.15 · Answer (C) I hope you will have washed all the clothes before you come into the house again.
The future perfect is 'will have' + past participle: 'will have washed'. 'Will have wash' uses the wrong form of the verb, and 'will be washing' is the future continuous.

Q.16 · Answer (C) Depresed
The correct spelling is 'depressed'.

Q.17 · Answer (A) bell the cat
To 'bell the cat' is to take on a dangerous task for the good of all: here, telling the disliked CEO what everyone feels.

Q.18 · Answer (B) An intrepid explorer stumbled upon an uncharted cavern deep within the heart of the rainforest.
SSC's key gives the sentence with 'an' for the explorer and the cavern, both met here for the first time, and 'the heart of the rainforest' (it has only one). The version with 'the' throughout is not ungrammatical, but it presumes an explorer and a cavern already known to the reader; 'a heart of the rainforest' is wrong.

Q.19 · Answer (D) C
A verb is modified by an adverb: 'She sings the song beautifully', so the error is in C.

Q.20 · Answer (C) for
One thanks someone 'for' something: 'thank you for the present'.

Q.21 · Answer (D) reflecting
'Mirrors reflecting the diverse facets of society': the '-ing' participle describes the mirrors.

Q.22 · Answer (D) for
A forum is a place 'for' discussion: 'a forum for dialogue, debate, and discussion'.

Q.23 · Answer (B) a
'Sense' here is a singular countable noun beginning with a consonant sound: 'fostering a sense of community'.

Q.24 · Answer (A) richness
After the adjective 'cultural' a noun is needed: 'highlight cultural richness'.

Q.25 · Answer (D) evolve
Newspapers continue to 'evolve', that is develop gradually, adapting to new mediums.

This paper online, with tap-to-reveal answers: https://parikshasaathi.com/exams/ssc-mts/question-papers/mts-2024-17-oct-2024-shift-3/
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